COMPSCI 4NL3 — Project Proposal

1) Team Members (Group 13)

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2) Task Title & Overview

Title

Turn-Level User Intent Classification for Task-Oriented Dialogues (MultiWOZ)

Overview

We will build a dataset and model that predicts the communicative intent of a user turn in a task-oriented conversation (e.g., finding a hotel, booking a train). This is useful for building dialogue systems because correctly detecting what the user is "doing" in the conversation (asking, giving constraints, confirming, correcting, etc.) is a key step before selecting the next system action.

Key Challenges

- Many turns look similar on the surface ("yes", "okay", "that works") but represent different intentions depending on context.
- Some turns combine multiple functions ("I need it cheap—any options downtown?"), which forces clear, consistent labeling rules.
- Some user utterances naturally express multiple intents within the same turn, which introduces ambiguity for single-label classification. Designing a labeling strategy that preserves the dominant intent while maintaining high inter-annotator agreement presents a non-trivial challenge.

Significance

- Correctly identifying user intent is a core component of task-oriented dialogue systems and NLUs.
- A well-designed intent classification dataset enables the development and evaluation of models that can robustly handle short, context-dependent utterances, which are common in real conversational interfaces such as virtual assistants and customer support bots.
- By constructing a dataset with carefully defined intent labels and explicit handling of multi-intent utterances, this project contributes a practical benchmark for studying intent recognition in realistic conversational settings, bridging the gap between simplified textbook examples and real-world dialogue behavior.

3) Task Definition

Type of data: English dialogue turns from task-oriented conversations.

Input:

- Context: previous system turn (one utterance)
- Current user utterance

Output:

Single-label multiclass classification (exactly one label per instance described below).

Label Set

We will annotate each user turn with one of these 5 labels:

- **REQUEST**: user asks for information/options or asks the system to do something (find, recommend, check availability).
- **INFORM_CONSTRAINT**: user provides preferences/constraints (price range, area, time, number of people, etc.).
- **CONFIRM_ACCEPT**: user accepts/approves a proposed option or confirms details ("yes book it", "that works", "correct").
- **CORRECT_CLARIFY**: user corrects the system or clarifies after misunderstanding ("no, not Tuesday, but Thursday", "I said 2 people").
- **SOCIAL**: greetings, thanks, closings, politeness-only ("thanks", "bye", "hello").

Single-label rule: if a turn has multiple functions, label the main purpose (we'll define tie-break rules in guidelines).

Examples

System: Would you like me to book the 6:15 train for you?
User: Yes, please book it for two people.
Assigned Label: CONFIRM_ACCEPT

System: I have booked a hotel for Friday night in the south area.
User: No, I said I needed it in the north.
Assigned Label: CORRECT_CLARIFY

4) Data Source & Data Collection Plan

Primary Data Source

We will use MultiWOZ (Multi-Domain Wizard-of-Oz), a widely used collection of ~10k and was created to support task-oriented dialogue modeling. Although there are many versions of this dataset, we will be using the most recent version MultiWOZ_2.2. This dataset has further refined dialogue state annotations.

Handling Existing Annotations

MultiWOZ contains various existing annotations (dialogue acts, states, etc.), but our project requires a task where we can annotate with reasonable agreement, so we will create our own label set and manually annotate the user turns using our guidelines (we will not "reuse" the provided labels as our gold labels). The course explicitly allows previously labeled data to be used as a source, but we will still do our own annotation work for our task.

Extraction & Preprocessing

Load dialogues and extract (previous system turn, current user turn) pairs. After this, we will keep only turns where:

- The current user turn length is within a reasonable range (e.g., 1 - 40 tokens) to avoid rare extremes.
- Normalize whitespace; keep punctuation (since it can carry meaning like questions).
- Store instances as CSV/JSON with: dialogue_id, turn_id, system_context, user_utterance, label.

Terms, Access, & Roadblocks

- This dataset has the MIT license, leaving everything to our discretion.
- No scraping required; dataset is directly downloadable and well-documented through standard research distributions.
- No rate limits and no API issues as it is an offline dataset.

5) Expected Dataset Size

Target Dataset Size

We plan to annotate 8,000 user turns total. The plan for annotation is as follows.

Annotation Effort Estimate

Estimated time per instance: ~15 seconds (short turn + one-turn context), with an estimated total labeling time of:

- 8,000 turns · 15 sec = $\sim$ 33.3 hours of labeling.
- With 3 teammates, that's $\sim$ 11 hours each, plus time for agreement labeling + adjudication.

Agreement Plan

- All members label the same 400-turn subset.
- Compute agreement (percent agreement + Cohen's kappa or Krippendorff's alpha).
- Resolve disagreements, refine guidelines, then label remaining data.

Example Annotation

(Format: System context → User turn → Label)

"I found several hotels in the north. Do you have a price range?"
→ "Cheap please, under 100 a night."
→ INFORM_CONSTRAINT

"Would you like me to book the 6:15 train?"
→ "Yes, book it for two people."
→ CONFIRM_ACCEPT

"What area would you like the restaurant in?"
→ "Do you have anything in the city centre?"
→ REQUEST

6) Team Contract

Team Name: Group 13
Course: COMPSCI 4NL3
Project: Turn-Level User Intent Classification (MultiWOZ)

Purpose & Mission

Our mission is to produce a high-quality, well-documented dataset and baseline models for a clear NLP task, by working consistently, communicating early about blockers, and holding each other accountable to deadlines and quality standards.

Expectations

- Attend a weekly meeting (30–45 minutes) and come prepared with progress updates.
- Respond in the group chat within 24 hours on weekdays.
- Do assigned work by internal deadlines (at least 48 hours before official due dates).
- Maintain quality by following annotation guidelines, logging uncertainties, and flagging confusing edge cases.

Roles & Duties

- **Coordinator (Varies Weekly):** Oversees project timeline, coordinates meetings, and ensures tasks are balanced.
- **Data Lead (Divij Dhiraaj):** Manages dataset extraction, preprocessing, and formatting (CSV/JSON).
- **Annotation Lead (Nicholas Zajkeskovic):** Develops and maintains the labeling guidelines, assigns turns for annotation and ensures inter-annotator agreement.
- **Modeling/Evaluation Lead (Mayowa Adesanya):** Implements baseline models for intent classification, designs evaluation metrics, runs experiments, and analyzes results.

Leadership & Facilitation

- We will rotate the coordinator each week.
- Decisions are made by consensus when possible; otherwise majority vote after discussion.
- Any major change to labels/guidelines requires notifying the whole team and updating the guideline doc.

Conflict & Accountability

- If someone is falling behind: bring it up early in chat and discuss in the next meeting, if needed we can also re-scope tasks.
- If issues persist we will schedule a 1:1 and bring a concrete plan to the group (what help is needed and what will be delivered by when).
- We will be respectful. The goal is fixing the process, not placing the blame on anyone.

Signatures

Print name	Signature	Date
Mayowa Adesanya		January 19, 2026
Divij Dhiraaj		January 19, 2026
Nicholas Zajkeskovic		January 19, 2026

7) References

MultiWOZ original paper (ACL 2018): `https://aclanthology.org/D18-1547/`
MultiWOZ 2.1 paper: `https://arxiv.org/abs/1907.01669`
Hugging Face distribution (multi_woz_v22): `https://huggingface.co/datasets/pfb30/multi_woz_v22`
GitHub source repo: `https://github.com/budzianowski/multiwoz`